

- Mevcut chatbot sistemleri, bir ifadenin sözcük anlamını başarılı bir şekilde işleyebilse de, iletişimin altında yatan sosyal ve bağlamsal kuralları, yani edimbilimi kavramakta zorlanmaktadır. Bu çalışmada bu durum "Edimbilimsel Boşluk" (Pragmatical Gap) olarak adlandırılmaktadır. Edimbilimsel Boşluk, iletişimsel ve sosyal etkileşimsel bir bağlamda gözlenen edimbilimsel özellik eksikliği olarak tanımlanabilir.

Thesis Title

Genuine: Pragmatical AI Fintech Chatbot for Digital Banking Applications

Thesis Paper Structure

- Giriş
- Projenin Kuramsal Temeli
- Edibilim Hakkında Bilgiler ve Açıklama (3 Teorinin Açıklaması)
- Bankacılık Dijital Asistanları Hakkında Bilgiler ve Açıklama
- Genuine Açıklaması (Teknik – Kod – Bilgiler)
- Veriler (Nicel-Nitel) – Figürler – İstatistiksel Grafikler
- Sonuç ve Bulgular
- Kaynakça (References)

Araştırma Sorusu

- Edimbilimsel dilbilim kuramları (Grice'ın İlkeleri, Söz Eylem Kuramı ve Nezaket Kuramı), dijital bankacılıktaki "Edimbilimsel Boşluk" (Pragmatic Gap) sorununu çözmek amacıyla Türkçe bir sanal asistan mimarisine (Genuine) hesaplamalı olarak nasıl entegre edilebilir?

Amaç ve Hedefler

Bu projenin amacı, edimbilimsel kuramları doğrudan temel alarak tasarlanmış, Türkçe dijital bankacılık uygulamalarına yönelik "Genuine" adlı yapay zeka destekli sanal asistanı tanıtmak ve mimari yapısını açıklamaktır.

Bu amaca ulaşmak için aşağıdaki hedefler belirlenmiştir:

- Söz Eylem, Grice ve Nezaket kuramlarını birleştirerek, durum izleme (state-tracking) ve RAG tabanlı bir edimbilimsel konuşma modeli oluşturmak.
- Açık kaynaklı Rasa ve GenAI altyapısını (Gemini API) kullanarak bankacılık senaryolarında (örn. yetersiz bakiye, kredi reddi) "Edimbilimsel Boşluğu" kapatan yazılımın teknik mimarisini belgelemek.
- Genuine'in yeteneklerini örnek diyalog akışları, ağ bağlantı görselleri (Gephi) ve edimbilimsel girdi-çıktı çözümlemeleri ile kısa, teknik ve odaklı bir sistem makalesi olarak sunmak.

Yöntem

Edimbilimsel Karşılıklı Konuşma Modeli Tasarımı

3 temel edimbilimsel kuramı birleştiren bir karşılıklı konuşma yönetim modeli tasarlanacaktır.

- Söz Eylem Tanımlama: Kullanıcının ifadesinin sadece yüzeysel niyetini değil, altında yatan asıl eylemi (Illocutionary act) anlamaya yönelik bir mantık geliştirilecektir. Örneğin, "Maaşımın bu kadar çabuk bittiğine inanmıyorum" ifadesi, bir şaşkınlık belirtisi olmasının yanı sıra "son harcamaları listele" talebi olarak yorumlanabilecektir.
- İşbirliğine Dayalı Karşılıklı Konuşma Yönetimi: Grice'ın dört ilkesi, sanal asistanın yanıt verme politikasını yönlendiren kurallar olarak formüle edilecektir. Örneğin, Nitelik İlkesi gereği, sanal asistan emin olmadığı bilgiler hakkında spekülasyon yapmayacak, Nicelik İlkesi gereği ise kullanıcıya sadece istenen bilgiyi, fazlasını veya eksikliğini vermeden sunacaktır.
- Nezaket Stratejisi Uygulaması: Özellikle kredi reddi, yetersiz bakiye gibi yüz tehdit edici eylemleri içeren durumlarda, Brown ve Levinson'ın Nezaket Kuramı'na dayalı yanıt stratejileri (örn. Olumsuz nezaket, dolaylı anlatım) tanımlanacaktır. Bu, "Yetersiz bakiye" gibi doğrudan bir yanıt yerine, "Görünüşe göre bu işlem için hesabınızda yeterli bakiye bulunmuyor. Dilerseniz mevcut bakiyenizi kontrol etmenize yardımcı olabilirim" gibi daha yumuşatılmış yanıtlar üretilmesini sağlayacaktır.

Makale/Kaynak Bilgileri

- Gephi** → Dilbilimsel birimlerin ilişkisini incelemek için Gephi uygulaması kullanılacak. Görsel ağ analizi için lider açık kaynaklı platformdur.
- Tableau/PowerBI** → Grafikler ve istatistiksel grafikler, modeller için kullanılacak.
- TrCOLA** → The original Corpus of Linguistic Acceptability consists of sentences compiled from English literature textbooks. The task is to determine if the sentences are grammatically correct and acceptable sentences. Our corpus is also compiled from Turkish linguistic textbooks and include morphological, syntactic and semantic violations. (Acceptability) (Corpus of Linguistic Acceptability)

A Grammar-based Dataset for Recovering Implicature and Conversational Reasoning

CITATION SECTION

1. Summary of the Paper

The paper, titled *GRICE: A Grammar-based Dataset for Recovering Implicature and Conversational rEasoning* (Zheng et al., 2021), addresses a critical flaw in modern natural language processing: AI models are great at understanding literal text but terrible at understanding what a human *actually means* (implicature).

To solve this, the authors created the "GRICE" dataset. This dataset maps multi-turn conversations where the user's statements contain hidden, implied meanings based on Grice's Maxims. They proved that even state-of-the-art conversational models fail at pragmatic reasoning compared to humans. However, when the authors added a specific module designed solely to "recover" or translate the implied meaning into an explicit statement before generating a response, the AI's conversational performance increased significantly.

2. Key Quotes and Information for Your Citations

You can use the following extracted quotes and data points directly in your thesis paper to provide academic backing for your hypotheses and methodology:

For Chapter 1 (Giriş) & Defining the "Pragmatic Gap":

"Understanding what we genuinely mean instead of what we literally say in conversations is challenging for both humans and machines; yet, this direction is mostly left untouched in modern open-ended dialogue systems." (Zheng et al., 2021).

- How to use it: Use this quote to introduce the core problem of your thesis. It proves that the "Pragmatic Gap" you are investigating is a recognized, unsolved challenge in the broader AI community.

For Chapter 2 (Kuramsal Temel) & Justifying RAG:

"To recover conversational implicature into explicit ones with only information and context in the dialogue, an ideal model should reason about the dialogue context and the relations among dialogue entities." (Zheng et al., 2021).

- How to use it: Use this to defend your decision to implement a RAG pipeline. It academically validates that to be pragmatically competent, an AI needs active reasoning over a specific context, not just static, rule-based training.

For Chapter 7 (Karşılaştırma Metodolojisi) & Hypothesis H1:

- Factual Citation: Zheng et al. (2021) demonstrated that baseline AI models experience a significant performance gap compared to humans in conversational reasoning. However, integrating a dedicated module that explicitly translates implied meanings into explicit statements yields an overall performance boost in conversational task success.
- How to use it: Use this to academically justify Hypothesis H1. It provides prior literature proving that adding a pragmatic reasoning layer (like Genuine) statistically outperforms a baseline model (like Finchat).

3. Advice on How to Approach This Paper

- For the Thesis Paper (The Architectural Blueprint): Use this paper to theoretically justify Genuine's RAG pipeline. Zheng et al. prove that simply processing literal text fails; models must explicitly "recover" the implied meaning. You can argue that Genuine's architecture operationalizes this exact finding by using a dedicated RAG "Pragmatic Guidebook" to explicitly reason over conversational implicatures before generating the final Turkish response.
- For the Coding Project (RAG Prompt Design): Adopt their 3-step "Literal -> Implicature -> Explicature" mapping to structure the JSON data in your system prompts. This provides a peer-reviewed, scientific schema for Genuine's few-shot examples.

Human-LLM Interaction Cycle

CITATION SECTION

1. Summary of the Paper

The paper, titled *Applying the Gricean Maxims to a Human-LLM Interaction Cycle: Design Insights from a Participatory Approach* (Kim et al., 2025), investigates how to make interactions with Large Language Models (LLMs) more cooperative and user-centered. The authors note that while LLMs are powerful, they lack the human ability to infer contextual nuances, which often leads to conversational breakdowns.

To solve this, the researchers conducted participatory design workshops with communication experts, interface designers, and frequent LLM users. The core outcome of this study is twofold: First, they reinterpreted Grice's original maxims specifically for human-LLM interactions (e.g., shifting the Maxim of Quality from just "truthfulness" to "fostering user trust" by showing reasoning). Second, they established nine actionable Design Considerations (DCs) distributed across three stages of interaction: communicating goals, generating outputs, and assessing outputs.

2. Key Quotes and Information for Citations

You can use the following extracted quotes and concepts directly in your thesis to justify both the problem you are solving and the prompt engineering strategies you use.

For Chapter 1 (Giriş) & Defining the Problem:

"One reason is that, unlike humans, LLMs lack inferential abilities that allow them to understand contextual nuances or interpret others' intentions."

- How to use it: Cite this when explaining why a baseline model like Finchat struggles with complex banking scenarios. It proves that the "Pragmatical Gap" is fundamentally an issue of inferential failure.

For Chapter 2 (Kuramsal Temel) & Justifying the Adaptation of Grice:

"Maxim of Quality, originally stressed providing truthful and evidence-based contributions, was expanded to include fostering user trust in the LLMs... This involves not only providing accurate information but also ensuring users can easily recognize its accuracy and reasoning process."

- How to use it: Use this to defend why Genuine's architecture includes specific conversational framing. You can argue that in a banking context, merely giving the right balance isn't enough; the chatbot must actively build trust through transparent communication.

For Chapter 6 (Genuine Açıklaması) & Justifying RAG/System Prompts:

- Factual Citation: Kim et al. (2025) propose specific design considerations for LLM generation, such as Design Consideration 7, which states that systems should enable the presetting of the LLM's role, characteristics, and response type to control interaction quality and depth.
- How to use it: Use this as the academic justification for your RAG "Pragmatic Guidebook." By feeding Gemini specific personas and response rules before it answers the user, you are directly implementing a peer-reviewed interaction design strategy.

3. Advice on How to Approach This Paper

- For the Thesis Paper (Prompt Engineering Justification): This paper is the bridge between 1970s Gricean theory and modern LLM mechanics. Use Section 4.1 to explain how Genuine translates abstract human conversational rules into machine-executable parameters.
- For the Coding Project (System Constraints): Use the 9 Design Considerations (DCs) from this paper to directly explain Genuine's generative constraints. For example, explain how Genuine satisfies "DC1: Quality" by algorithmically providing the banking policy reason before stating a loan rejection, ensuring the software builds user trust.

Bankacılıkta Dijital Dönüşümle Değişen Müşteri Deneyimi

CITATION SECTION

1. Summary of the Paper

The paper, titled *Bankacılıkta Dijital Dönüşümle Değişen Müşteri Deneyimi: Müşteri Sadakati, Memnuniyeti ve Tavsiye Eğilimine Yansımaları* (Aydın & Onaylı, 2020), investigates how the rapid shift toward digital banking in Turkey affects customer attitudes.

The authors conducted a quantitative survey of 364 Turkish digital banking customers and analyzed the data using the PLS-SEM method. The study reveals a fascinating market reality: Turkish customers no longer differentiate between banks based on transaction speed, security, customer support, or information content. Because regulations are strict and technology is widely available, customers view these features as standard expectations rather than competitive advantages.

Instead, the study found that the only two factors that significantly drive customer satisfaction and loyalty are usability (kullanışlılık) and ease of use (kullanım kolaylığı). Furthermore, when customers are satisfied and loyal, they are highly likely to recommend the bank to others. The paper also documents the massive migration of Turkish users from traditional internet banking to mobile-only banking.

2. Key Quotes and Information for Citations

You can extract the following data points and quotes to build strong, localized arguments in your thesis:

For Chapter 1 (Giriş) & The Problem Statement:

- The Loss of Human Touch: *"Dijital uygulamalar her ne kadar hızlı ve kolay süreçler getiriyor olsa da müşteri ve banka çalışanı arasındaki birebir insan ilişkisini azaltmaktadır".*
 - How to use it: Use this to define the problem. Digital banking is fast, but it feels robotic. This perfectly sets up your argument that Genuine's pragmatic framework (Politeness and Grice's Maxims) restores that lost "human relationship" in a digital space.
- The Shift to Mobile: As of March 2020, the number of customers using *only* mobile banking in Turkey was 11 times higher than those using only internet banking (44 million vs 3.8 million).
 - How to use it: Cite this statistic to justify why optimizing conversational UI on small screens (via chatbots) is the most critical frontier for Turkish banks right now.

For Chapter 4 (Bankacılık Chatbot'ları) & Justifying Genuine:

- The Only Differentiators: *"Araştırmada dikkat çeken sonuç sunulan dijital uygulamaların müşteriler tarafından kullanım kolaylığı ve kullanışlılığına yapılan vurgudur. Zira yapılan işlemlerde müşteri tatmini ve müşterinin bankaya sadakati bu iki özellik tarafından belirlenmektedir".*

- *How to use it:* This is your golden quote. Use it to prove that "Ease of Use" is the most valuable metric in Turkish banking. You can then argue that a bot that forces users to speak in rigid keywords (Finchat) violates "ease of use," while a bot that understands natural, implied meaning (Genuine) maximizes it.
- The Need for Chatbots: *"Dolayısıyla, bankaların dijital uygulamalarındaki menüleri kullanıcı dostu bir şekilde tasarlamaları, aranan işlemin hızlı bir şekilde bulunmasını sağlamaları önemlidir. Birçok banka tarafından son zamanlarda tanıtılan sanal asistanlar da bu amaca hizmet etmektedir".*
 - *How to use it:* Use this to connect the concept of "usability" directly to the necessity of virtual assistants in the banking sector.

3. Advice on How to Approach This Paper

- For the Thesis Paper (The Market Necessity): Use this to answer *why* Genuine was built. According to Turkish market data, standard features like speed and security are no longer competitive advantages; the sole differentiators are "usability" and "ease of use." You can frame Genuine as a direct software solution to this market demand, as its pragmatic architecture (understanding natural, implied meaning instead of rigid keywords) maximizes usability for the Turkish digital banking sector.

Beyond Understanding: Evaluating the Pragmatic Gap in LLMs

CITATION SECTION

1. Summary of the Paper

The paper, titled *Beyond Understanding: Evaluating the Pragmatic Gap in LLMs' Cultural Processing of Figurative Language* (Attia et al., 2026), systematically evaluates 22 different LLMs on their ability to process culturally grounded figurative language (idioms and proverbs).

The authors used figurative language as a testing ground to see if LLMs actually possess cultural and pragmatic competence. They tested the models across three tasks: contextual understanding, pragmatic use, and interpretation of connotations.

The results revealed a massive "Pragmatic Gap." While LLMs were reasonably okay at basic contextual understanding, their accuracy plummeted by 14.07% when asked to *pragmatically use* the language. Furthermore, the study exposed a severe language bias: models performed significantly worse on Arabic proverbs compared to English, and even worse on localized Egyptian Arabic. However, the study also found a silver lining: providing the LLM with contextual sentences improved pragmatic accuracy by 10.66%.

2. Key Quotes and Information for Citations

You can extract the following data points to build strong, empirical arguments in your thesis:

For Chapter 1 (Giriş) & Defining the "Pragmatic Gap":

- The Core Definition: Attia et al. (2026) demonstrate that there is a measurable discrepancy—termed the "Pragmatic Gap"—between an LLM's superficial understanding of text and its ability to appropriately use and interpret deep connotative meanings.
 - *How to use it:* Use this to formally define what the "Edimbilimsel Boşluk" is in your introduction. It proves your thesis topic is a recognized, cutting-edge issue in modern NLP.

For Chapter 2 (Kuramsal Temel) & Justifying the Focus on Turkish:

- The Language Bias: The study revealed a consistent hierarchy of failure in LLMs: accuracy for Arabic proverbs was 4.29% lower than English, and Egyptian dialects were 10.28% lower than standard Arabic.
 - *How to use it:* Use this data to justify why your research on *Turkish* banking chatbots is necessary. If LLMs fail at Arabic pragmatics compared to English, they inherently struggle with Turkish pragmatics as well. This proves that out-of-the-box Gemini/ChatGPT is not enough for the Turkish market without your custom architecture.

For Chapter 6 (Genuine Açıklaması) & Justifying the RAG Architecture:

- The Contextual Fix: The authors found that "providing contextual idiomatic sentences improves the accuracy by 10.66%".
 - *How to use it:* This is the academic justification for your entire RAG "Pragmatic Guidebook." You can state: *Attia et al. (2026) proved that injecting explicit contextual rules improves an LLM's pragmatic accuracy by over 10%; therefore, Genuine utilizes a RAG pipeline to dynamically supply Gricean and Politeness contexts to the generative model.*

3. Advice on How to Approach This Paper

- For the Thesis Paper (Defining the Software's Purpose): Treat this paper as the definition of the exact technical bug Genuine patches. Use it to establish that off-the-shelf LLMs possess a measurable "Pragmatic Gap," especially in non-English languages like Arabic (and by extension, Turkish). Genuine's entire software architecture is positioned as a localized middleware solution that injects the missing cultural and pragmatic competence into the LLM cycle.

Evaluating Dialogs based on Grice's Maxims

CITATION SECTION

1. Summary of the Paper

The paper, titled *Evaluating Dialogs based on Grice's Maxims* (Jwalapuram, 2017), addresses the lack of a standardized, effective evaluation metric for conversational dialog systems. The author argues that traditional objective metrics (like task completion time) and subjective metrics (like overall user satisfaction surveys) often fail to accurately reflect the true quality of the interactive experience.

To solve this, the study proposes using Grice's Maxims as a structured evaluation framework. The researcher collected 11 system-generated dialogs (a mix of task-oriented and conversational bots) and had 62 participants rate them on a 1-5 Likert scale based on four specific questions mapping to Quantity, Quality, Relation (Relevance), and Manner.

The results proved that evaluating dialogue through this Gricean lens is highly effective. Task-oriented dialogs generally scored higher than conversational ones. Notably, the study found that systems using *implicit* confirmation (understanding the user and moving forward) scored significantly higher in Relevance and Manner than systems using *explicit* confirmation (constantly asking "Yes or No?"), even when both accomplished the same task. Ultimately, the paper demonstrates that mean scores derived from Gricean maxims can serve as a reliable threshold to distinguish between acceptable and unacceptable dialog systems.

2. Key Quotes and Information for Citations

You can extract the following data points to build strong, theoretically grounded arguments in the methodology and design chapters:

For Chapter 7 (Karşılaştırma Metodolojisi) & Justifying the Evaluation Metric:

- The Flaw in Traditional Metrics: "In the case of task-oriented systems, objective metrics such as dialog success rate or completion time do not always correspond to the most effective user experience due to the interactive nature of dialog".
 - *How to use it:* Cite this to justify why a simple task-completion metric is insufficient for evaluating banking assistants, thereby proving the necessity of measuring the "Pragmatic Gap" using Gricean principles.

For Defining the Scoring Rubric:

- The 4-Point Evaluation Framework: The paper establishes four specific questions to quantify the maxims:
 - Quantity: "Is the system as informative as it can be and gives as much information as is needed and no more?".
 - Quality: "Is the system truthful or does it give information that is false, to the best of your world knowledge?".
 - Relation: "Are the system responses relevant, i.e., does the system respond with things pertinent to the discussion?".
 - Manner: "Are the system responses clear, orderly and without obscurity and ambiguity?".
 - *How to use it:* Cite Section 3.2 of this paper as the direct academic source for the exact rubric parameters used in the comparative analysis.

For Chapter 6 (Genuine Açıklaması) & Dialogue Management Design:

- Implicit vs. Explicit Confirmation: The study compares Dialog 5 (implicit confirmation) and Dialog 6 (explicit confirmation), noting that Dialog 5 performs significantly better across all counts, with a massive difference in relevance and manner ratings. "This indicates that users do not like explicit confirmations despite them having a higher task success rate...".
 - *How to use it:* Use this finding to theoretically defend the architectural choices made in the Rasa dialogue management configuration, specifically minimizing repetitive "Yes/No" verification loops in favor of smoother, implicit context tracking.

3. Advice on How to Approach This Paper

- For the Thesis Paper (Dialogue Flow Design): Since you are no longer doing A/B testing with a human/LLM rubric, pivot the use of this paper to explain Genuine's *dialogue management design*. Jwalapuram found that systems using "implicit confirmation" score significantly higher in Relation and Manner than those using rigid "explicit confirmation." Use this to document why Genuine's Rasa state-tracker is engineered to move the conversation forward implicitly, reducing frustrating "Yes/No" loops.

Factors affecting chatbots in banking services

CITATION SECTION

1. Summary of the Paper

The paper, titled *Factors affecting chatbots in banking services: the UTAUT2 and innovation resistance theory perspective* (Marak et al., 2025), investigates what drives or prevents users from adopting banking chatbots in an emerging economy (India).

The researchers combined two established frameworks: the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) and Innovation Resistance Theory (IRT). Using a survey of 194 digital banking customers and analyzing the data with PLS-SEM, they found that Performance Expectancy, Hedonic Motivation, and Social Influence positively drive chatbot adoption, while the Risk Barrier significantly hinders it.

The study also conducted a fascinating sub-sample analysis based on age. It revealed that Gen Z users are heavily driven by "Hedonic Motivation" (seeking an enjoyable, engaging experience), whereas older demographics are more influenced by "Social

Influence" (seeing others use it). Both age groups, however, shared a strong concern regarding the "Risk Barrier," indicating that a lack of trust severely impacts the willingness to use banking bots.

2. Key Quotes and Information for Citations

You can extract the following data points to build strong, market-driven arguments in your thesis:

For Chapter 1 (Giriş) & The Problem Statement:

- The High Stakes of Banking Bots: "The rise of chatbot technology in the banking sector has been notable, with banks increasingly deploying these systems to enhance service delivery".
 - *How to use it:* Cite this to establish that while chatbot deployment is widespread, ensuring their conversational usability is critical for success.

For Chapter 4 (Bankacılık Chatbot'ları) & Justifying Genuine:

- Performance Expectancy: "When users perceive chatbots as efficient tools for swiftly obtaining comprehensive banking information and services, it creates a positive attitude and contribute to their inclination to engage...".
 - *How to use it:* Use this to argue that a bot must do more than just route keywords. It must efficiently handle the underlying intent (Speech Acts) to fulfill performance expectations.
- The Risk Barrier (Trust): "As individuals perceive risks to their privacy, confidentiality, and security, their trust gets affected, and hence, they will be less willing to use such services".
 - *How to use it:* Use this quote to bridge the gap between technical design and user adoption. You can argue that blunt, uncooperative responses from baseline models trigger these risk barriers by making the bot feel alien or unreliable, whereas pragmatically competent models actively build trust.
- Hedonic Motivation for Younger Users: "When users perceive using chatbots as not only functional but also as a source of enjoyment, fun, and entertainment, it is likely to enhance their intention...".
 - *How to use it:* Cite this to justify why making a bot "cooperative" (Grice's Maxims) and "polite" (Brown & Levinson) creates the engaging, positive user experience that modern users demand.

3. Advice on How to Approach This Paper

- For the Thesis Paper (Handling Edge Cases): Use this paper to explain Genuine's error-handling and edge-case logic. Marak et al. identify the "Risk Barrier" as the primary reason users abandon banking bots. You can document how Genuine's integration of Politeness Theory specifically triggers during high-risk conversational states (e.g., insufficient funds) to algorithmically lower the user's perceived risk and prevent session abandonment.

Pragmatics and Computational Linguistics

CITATION SECTION

1. Summary of the Paper

The paper, *Pragmatics and Computational Linguistics* (Jurafsky, 2006), explores how computational systems attempt to understand the relationship between a user's utterance and the surrounding context. Jurafsky specifically focuses on the problem of "speech act interpretation"—how a machine figures out what a user *means* to do (e.g., request, confirm, inform) rather than just what they literally say.

To solve this, the paper outlines two dominant computational paradigms:

- The Plan Inference (BDI - Belief, Desire, Intention) Model: This is a logic-based approach that assumes an utterance has a literal meaning, which the system then passes through a chain of logical "plan inference rules" to deduce the hidden, indirect meaning. For example, determining that a question about ability ("Can you give me a list of flights?") is actually an indirect request. While deep and explanatory, this model struggles to scale.
- The Cue-Based (Probabilistic) Model: This approach abandons the idea of a "literal" meaning. Instead, it treats interpretation as a statistical classification task, relying on surface "cues" such as lexical markers (specific words), syntax, prosody (intonation), and conversational structure to predict the most likely dialogue act.

The paper also highlights the evolution from classical philosophical "Speech Acts" to more granular, machine-friendly "Dialogue Acts" (like BACKCHANNELS, CHECKS, and REFORMULATIONS) used in actual conversational AI engineering.

2. Key Quotes and Information for Citations

You can extract the following concepts and quotes to build the theoretical bridge between linguistics and software engineering in your thesis:

For Chapter 3 (Edimbilim) & Defining the Field:

- The Core Definition: "Computational pragmatics might be defined as the computational study of the relation between utterances and context".
- The Role of Inference: "...computational pragmatics, like pragmatics in general, is especially concerned with INFERENCE".
 - *How to use it:* Use these to introduce the specific sub-field your thesis occupies, establishing that interpreting pragmatics requires active computational inference, not just static keyword matching.

For Chapter 2 (Kuramsal Temel) & The Flaw of Literal Interpretation:

- Rejecting Literal Force: "...the speech act force of most utterances does not match their surface form".
- The Need for Deep Reasoning: To interpret an indirect request, a system requires an inference chain that maps the literal text (e.g., a question about the system's ability) to the user's actual intention (the request for action).
 - *How to use it:* Cite this when explaining why rule-based bots fail at complex banking scenarios. They get stuck on the "Literal Force" (e.g., assuming a user expressing stress about their salary is just making a statement, rather than requesting a transaction review).

For Chapter 6 (Genuine Açıklaması) & Defining Intents:

- Dialogue Acts: Modern computational work has expanded classical speech acts into "dialogue acts" to model the granular conversational functions an utterance performs in a system.
 - *How to use it:* Use this to justify the specific intent categories created within the chatbot's NLU pipeline.

3. Advice on How to Approach This Paper

- For the Thesis Paper (The Hybrid Engine): Use Jurafsky to explain Genuine's underlying technical philosophy. Genuine is a hybrid engine: it uses the "Cue-Based" model (Rasa's statistical NLU) to detect the initial intent, and the "Plan Inference" model (Generative AI + Pragmatic Guidebook) to execute the deep, context-aware reasoning. This categorizes your software precisely within the historical taxonomy of computational linguistics.

The ABC of Computational Pragmatics

CITATION SECTION

1. Summary of the Paper

The paper *The ABC of Computational Pragmatics* (Bunt & Black, 2000) serves as a foundational text that bridges abstract linguistic theory with the engineering of Natural Language Processing (NLP) systems. The "ABC" in the title stands for Abduction, Belief, and Context—the three pillars the authors argue are necessary for a machine to truly understand human language.

The paper argues that processing literal meaning is entirely insufficient because natural language is heavily "indexical" (context-dependent). To resolve this, the authors focus heavily on Dialogue Act Modeling and Context Updating. They propose that a conversation is not just an exchange of sentences, but a continuous process where each utterance serves as an "update" to a shared mental state between the speaker and the hearer. To understand what a user means, a computational system must use abductive reasoning—working backward from the utterance to infer the user's underlying beliefs and intentions.

2. Key Quotes and Information for Citations

You can extract the following concepts and quotes to build a rigorous theoretical defense for your dialogue management architecture:

For Chapter 2 (Kuramsal Temel) & Defining Computational Pragmatics:

- The Core Definition: "Pragmatics is the branch of linguistics that studies the relations between linguistic phenomena and aspects of the context of language use".
 - *How to use it:* Use this as your opening definition in your theory chapter to establish the exact scientific boundaries of your research.
- The Necessity of Context: "The information partially encoded in the linguistic expression must become fully specified relative to an actual context in order for the expression to have a determinate meaning".
 - *How to use it:* Cite this to explain *why* banking chatbots fail when they only use keyword matching. A statement like "I just got paid" only has a "determinate meaning" if the system understands the banking context (e.g., the user wants to check if a specific deposit cleared).

For Chapter 3 (Edimbilim) & Abductive Reasoning:

- The Role of Abduction: Bunt and Black detail that understanding a user's true intent requires abductive reasoning—inferring the most plausible explanation for an utterance based on the current conversational state.
 - *How to use it:* Use this to academically justify why a simple rule-based system (Finchat) struggles with the Pragmatic Gap. Finchat operates on deterministic deduction, whereas interpreting human speech acts requires dynamic abduction.

3. Advice on How to Approach This Paper

- For the Thesis Paper (State Tracking Mechanics): Use Bunt and Black's theory of "Context Updating" as the academic terminology for Genuine's backend memory. When detailing the technical pipeline, explain that Genuine does not process isolated sentences; rather, it uses Rasa's slot-filling to perform continuous "Context Updating," allowing the RAG layer to access a fully specified, dynamic conversational state.

The Gricean Maxims in NLP - A Survey

CITATION SECTION

1. Summary of the Paper

The paper, titled *The Gricean Maxims in NLP - A Survey* (Krause & Vossen, 2024), is the first comprehensive review of how Grice's cooperative principles have been utilized to develop and evaluate Natural Language Processing (NLP) systems.

The authors systematically explore the application of the maxims (Quantity, Quality, Relation, and Manner) across various NLP domains, including Natural Language Generation (NLG), Natural Language Understanding (NLU), and Interactive Systems like conversational agents. A major focus is placed on recent Large Language Models (LLMs), revealing that while top-tier models like GPT-4 perform well, they still struggle significantly with pragmatic nuances—particularly by overgenerating information (violating Quantity) and failing to grasp culturally specific or violated social norms. The paper also highlights an important critique: the maxims were designed as universal, but real-world application often requires *cultural adaptation*, as different societies interpret these conversational rules differently.

2. Key Quotes and Information for Citations

This paper is a goldmine because it aggregates decades of research. You can use it to justify almost every major architectural and evaluative decision in your thesis.

For Chapter 2 (Kuramsal Temel) & Justifying the Maxims in AI:

- The Foundational Value: "Originating from the domain of pragmatics, the Gricean maxims are foundational principles aimed at optimising communicative effectiveness...".
 - *How to use it:* Cite this to establish that Grice's theories are not just abstract philosophy, but are actively recognized as the standard for optimizing modern AI communication.

For Chapter 4 (Bankacılık Chatbot'ları) & The Cost of the Pragmatic Gap:

- User Frustration: The survey highlights research demonstrating that "...violations of Grice's maxims in interactions with Alexa led to user frustration, with Relevance violations being particularly aggravating". Furthermore, these violations increase the "cognitive load" on the user, significantly burdening the interaction process.
 - *How to use it:* Use this to prove the business and user-experience costs of the "Pragmatic Gap." A bot like Finchat that gives rigid or irrelevant responses literally forces the user to work harder to understand the system.

For Chapter 7 (Karşılaştırma Metodolojisi) & Validating Your LLM Judge:

- Evaluation Standard: "Evaluating conversational agents for their adherence to Gricean maxims provides insights into their effectiveness and user satisfaction". The survey notes that numerous works "...propose frameworks where human raters assess dialogues based on Gricean categories".
 - *How to use it:* This is your ultimate defense for your A/B testing rubric. You can argue that evaluating chatbots strictly on Quantity, Quality, Relation, and Manner is a heavily vetted, state-of-the-art methodology in NLP.

For Chapter 10 (Sonuç/Tartışma) & The Turkish Context:

- Cultural Limitations: "...certain cultural settings might interpret the maxims differently, indicating a need for cultural adaptation of these principles".
 - *How to use it:* Use this to emphasize the value of your specific focus on *Turkish* banking. You can argue that applying Gricean maxims to Turkish fintech requires localized adjustments (which you handled via Brown & Levinson's politeness markers), adding unique value to the global literature.

3. Advice on How to Approach This Paper

- For the Thesis Paper (State-of-the-Art Validation): As a comprehensive 2024 survey, this paper is your definitive proof that Genuine utilizes cutting-edge methodologies. Use it in the introduction to state that embedding Gricean maxims into LLM architectures is the current frontier for mitigating generative over-saturation (violating Quantity) and ensuring safe, context-aware conversational AI.

Unleashing the Power of Smart Chatbots: Transforming Banking with Artificial Intelligence

CITATION SECTION

1. Summary of the Paper

The paper *Unleashing the Power of Smart Chatbots: Transforming Banking with Artificial Intelligence* (Balaji et al., 2024) investigates the impact of AI-driven chatbots on customer satisfaction and operational efficiency within the banking sector. The authors conducted an empirical study using a structured questionnaire distributed to 285 chatbot users, analyzing the data via partial least squares structural equation modeling (PLS-SEM).

The core finding is that smart chatbots have a significant positive impact on customer satisfaction by providing personalized and timely assistance. More specifically, the study mathematically proves that a customer's behavioral intention to use a banking chatbot is heavily influenced by five distinct factors: Ease of Use, Usefulness, Enjoyment, Innovativeness, and Information Quality. The paper also provides a high-level architectural overview of modern chatbots, breaking them down into components like the User Interface, Natural Language Processing (NLP), Dialog Management, and Knowledge Bases.

2. Key Quotes and Information for Citations

You can extract the following datapoints to build strong, market-driven arguments in your thesis:

For Chapter 1 (Giriş) & Justifying the Research:

- The Core Premise: "With the rise of digital transformation and the increasing demand for seamless customer experiences, banks are leveraging smart chatbots to enhance customer interactions and improve operational efficiency".
 - *How to use it:* Use this to set the stage in your introduction. It establishes that conversational AI is not just a novelty, but a core operational necessity for modern banks.

For Chapter 4 (Bankacılık Chatbot'ları Hakkında Bilgiler) & The Five Adoption Factors:

- The Drivers of Customer Behavior: The results of the structural modeling highlight that "Customer Behavior (B) is significantly impacted by Ease of Use (E), Usefulness (U), Innovation (I), Enjoyment (E), and Information Quality (Q)".
 - *How to use it:* This is your business justification. You can argue that while traditional bots check the box for "Usefulness" by performing basic tasks, they often fail at "Enjoyment" and "Information Quality" due to rigid, uncooperative responses.

For Chapter 5 (Finchat Açıklaması) & Defining the Architecture:

- Chatbot Anatomy: "Dialog Management: Dialog management controls the flow of conversation between the chatbot and the user".
 - *How to use it:* When describing Finchat's baseline architecture, cite this to academically define the role that the Rasa framework plays (handling dialog management and NLP) before you introduce the generative RAG enhancements in the next chapter.

3. Advice on How to Approach This Paper

- For the Thesis Paper (System Context & Architecture): Use this to outline the standard anatomy of a banking bot (NLP, Dialog Management, Knowledge Base) before introducing how Genuine *modifies* this standard anatomy. It grounds your software in established enterprise architecture, allowing you to highlight Genuine's specific pragmatic enhancements (the RAG Guidebook) over the typical industry baseline.

